

MATH 180 Strategies of Problem Solving

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Outline

1 SoPS Lecture 8

- Last time
- Sprouts
- Graph theory
- Sprout puzzles
- Wrapping up

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Review

Last time

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- we played Break the Cube

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- we played Break the Cube
- we talked about proofs

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Last time

- we played Break the Cube
- we talked about proofs
- we constructed a Break the Cube: Solitaire puzzle

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Sprouts by John Conway

Rules

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- draw a few starting seeds

Sprouts by John Conway

Rules

- draw a few starting seeds
- each turn a player must
 - **sprout** a vine between two seeds
 - **spawn** a new seed along the vine
- restrictions
 - **no overlaps**: vines cannot cross
 - **no overflow**: seeds can have at most 3 vines attached

Sprouts by John Conway

Rules

- draw a few starting seeds
- each turn a player must
 - **sprout** a vine between two seeds
 - **spawn** a new seed along the vine
- restrictions
 - **no overlaps**: vines cannot cross
 - **no overflow**: seeds can have at most 3 vines attached
- first player unable to sprout loses

Two-seed game example

START

Two seeds, no lines yet



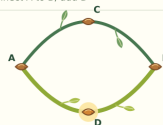
TURN 1

Connect A to B; add C



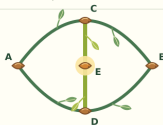
TURN 2

Connect A to B; add D



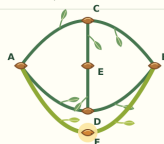
TURN 3

Connect C to D; add E



TURN 4

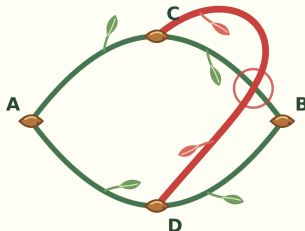
Connect A to B; add F



No overlaps

ILLEGAL: crossing vines

Attempt: connect C to D along the red vine

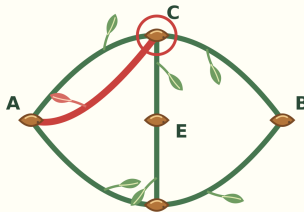


The red vine crosses an existing green vine.

No overflow

ILLEGAL: fourth vine

Attempt: connect C to A along the red vine



C already touches three vines.

Let's play!

Activity

Let's play!

Activity

- play a game of sprouts with 3 – 6 starting seeds

Let's play!

Activity

- play a game of sprouts with 3 – 6 starting seeds
- **record**: number of starting seeds, number of seeds at end

Let's play!

Activity

- play a game of sprouts with 3 – 6 starting seeds
- **record**: number of starting seeds, number of seeds at end
- think of an interesting question to ask about the game

Questions

Questions

Question 1

Can you tell from the final number of seeds how many turns were taken?

Questions

Question 1

Can you tell from the final number of seeds how many turns were taken?

Question 2

Does the game always have to end?

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Graphs

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Definition

A **graph** or **network** is a collection of points called **vertices** or **nodes** connected together by curve segments that we call **edges**.

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- sprout games make graphs!
- social networks

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- sprout games make graphs!
- social networks
- linked pages on Wikipedia

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Examples:

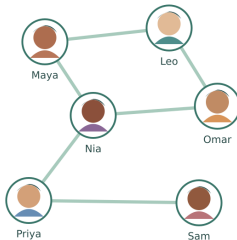
- sprout games make graphs!
- social networks
- linked pages on Wikipedia
- aviation/flight networks

Social network

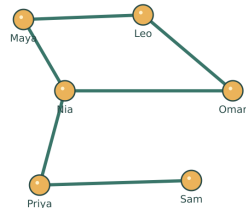
A social network

Person $\rightarrow$ vertex • friendship $\rightarrow$ edge

PEOPLE



THE GRAPH



Wikipedia articles

Wikipedia hyperlinks

Article $\rightarrow$ vertex • hyperlink $\rightarrow$ directed edge

ARTICLE PAGES

•

Graph theory

[Seven Bridges](#)
[Leonhard Euler](#)
[Topology](#)

•

Seven Bridges of Königsberg

[Graph theory](#)
[Eulerian path](#)

•

Leonhard Euler

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Eulerian path

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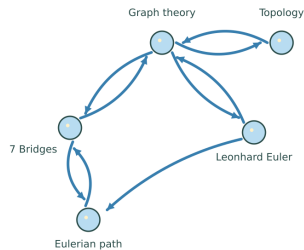
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Topology

[Graph theory](#)



THE DIRECTED GRAPH



Direct flights

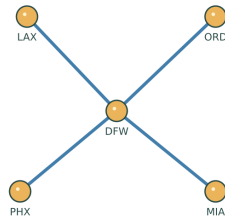
Direct flights between U.S. cities

Airport → vertex • nonstop route → edge

AIRPORTS ON A MAP



THE SAME NETWORK

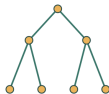


Zoo of graphs

A small zoo of graphs

Dots are vertices; each line records a connection between two vertices.

Tree



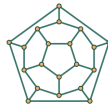
7 vertices - 6 edges

Square lattice



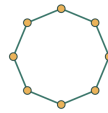
16 vertices - 24 edges

Dodecahedral graph



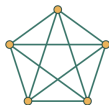
20 vertices - 30 edges

Cycle C_8



8 vertices - 8 edges

Complete graph K_5



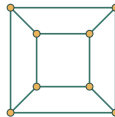
5 vertices - 10 edges

Petersen graph



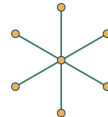
10 vertices - 15 edges

Cube graph Q_3



8 vertices - 12 edges

Star $K_{1,6}$



7 vertices - 6 edges

Properties of sprout graphs

Properties of sprout graphs

Question

Can we tell if a particular graph came from a game of Sprouts?
What properties should it have?

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The **degree** of a vertex is how many edges are coming out of it.

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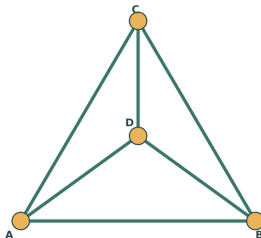
A graph is called **planar** if no edges in the graph overlap.

Planar graphs

Planar and nonplanar graphs

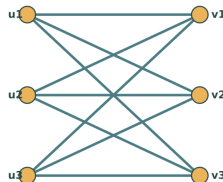
Planar means that some drawing has no crossing edges.

PLANAR: K_4



All six edges fit without a crossing.

NONPLANAR: $K_{3,3}$

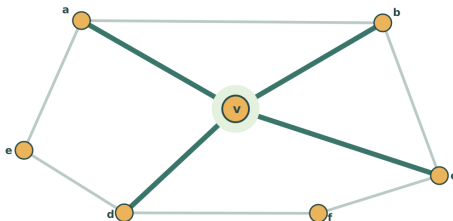


Every drawing of $K_{3,3}$ has a crossing.

Vertex degree

The degree of a vertex

Count the edges that touch the highlighted vertex.



$\deg(v) = 4$ four highlighted edges meet at v

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Longest game

Activity:

For each positive integer n , let S_n denote the maximum number of seeds you can create playing Sprout starting with n seeds. Working with your group, determine the value of S_1 , S_2 , S_3 , S_4 , and S_5 .

Problem:

Can you see a pattern?

Shortest game

Activity:

For each positive integer n , let M_n denote the minimum number of seeds you can create playing Sprout starting with n seeds. Working with your group, determine the value of M_1 , M_2 , M_3 , M_4 , and M_5 .

Problem:

Can you see a pattern?

Separate regions

Activity:

For the various games you played, can you find a relationship between the starting number of seeds, the number of turns that have been played, and the number of different regions the graph cuts the plane into?

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Question of the Day

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Puzzle

Suppose you are in a room with five other people. Explain why there must be a group of three people who all know each other or a group of three people none of which know each other.

Final thoughts

Reflection

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- Describe what a graph is and find an example from real life of some information that can be represented by a graph.

Final thoughts

Reflection

- Describe what a graph is and find an example from real life of some information that can be represented by a graph.
- Look up a cool fact or theorem about graphs online and describe it in your own words.

Submit your responses by **Tonight!**